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WGU MSDA D212

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Task 3

Part I: Research Question

A.

1. Can we use Market Basket Analysis to identify customer purchasing association behavior?

2. The goal is to use Market Basket Analysis (MBA) to identify purchasing associations between items made by customers. This will be useful for business-making decisions and further research and analysis.

Part II: Market Basket Justification

B.

1. Market Basket Analysis (MBA) is a data mining technique that finds patterns of purchases that are frequently paired/combined together. An obvious example would be hot dogs and hot dog buns.

It starts with transaction data, usually structured as rows representing purchases and columns representing items that could appear in that purchase. Then, using an algorithm like Apriori, it scans through the dataset and identifies frequent combinations of items that appear together more frequently than expected at baseline chance. It then generates "if-then rules", which are structured like: "If a person buys hot dogs, they are likely to also buy hot dog buns."

These rules are evaluated using 3 metrics: Support, Confidence, and Lift. Support is a measure of how often an itemset appears in a transaction table. Confidence is a measure of how often an item is purchased assuming that an specific initial item has also been purchased. Lift is a measure of how often items would appear together if they are adjusted for how often the first item appears relative to other items (Doe).

2. An example transaction in this dataset would be the first transaction in the dataset (Row 5). They are listed as having purchased: Apple Lightning to Digital AV Adapter, TP-Link AC1750 Smart WiFi Router, and Apple Pencil. This means that these three items were purchased by a customer in the same transaction.

3. One assumption of MBA is that the data is representative of true underlying associations rather than external or temporary trends, such as sales, promotions, or other such events.

Part III: Data Preparation and Analysis

C.

1. **Cleaned dataset attached.**

2. Full code is attached, however here are some screenshots:

```
#Apriori time
#Code sourced from either Dr. Kamara's videos or
#https://rasbt.github.io/mlxtend/USER_GUIDE_INDEX/
from mlxtend.frequent_patterns import apriori
from mlxtend.frequent_patterns import association_rules
rules = apriori(df4, min_support = 0.02, use_colnames = True)
rules.head(5)
```

	support	itemsets
0	0.050527	(10ft iPHone Charger Cable 2 Pack)
1	0.042528	(3A USB Type C Cable 3 pack 6FT)
2	0.029463	(Anker 2-in-1 USB Card Reader)
3	0.068391	(Anker USB C to HDMI Adapter)
4	0.087188	(Apple Lightning to Digital AV Adapter)

```
#Rules table
rulestable = association_rules(rules, metric = 'lift', min_threshold = 1)
rulestable.head(10)
```

	antecedents	consequents	antecedent support	consequent support	support	confidence	lift	representativity	leverage	conviction	zhangs_metric	jaccard	certainty	kulczynski
0	(10ft iPHone Charger Cable 2 Pack)	(Dust-Off Compressed Gas 2 pack)	0.050527	0.238368	0.023064	0.456464	1.914955	1.0	0.011020	1.401255	0.503221	0.086760	0.286354	0.276610
1	(Dust-Off Compressed Gas 2 pack)	(10ft iPHone Charger Cable 2 Pack)	0.238368	0.050527	0.023064	0.096756	1.914955	1.0	0.011020	1.051182	0.627330	0.086760	0.048690	0.276610
2	(Dust-Off Compressed Gas 2 pack)	(Anker USB C to HDMI Adapter)	0.238368	0.068391	0.024397	0.102349	1.496530	1.0	0.008095	1.037830	0.435627	0.086402	0.036451	0.229537
3	(Anker USB C to HDMI Adapter)	(Dust-Off Compressed Gas 2 pack)	0.068391	0.238368	0.024397	0.356725	1.496530	1.0	0.008095	1.183991	0.356144	0.086402	0.155399	0.229537
4	(VIVO Dual LCD Monitor Desk mount)	(Anker USB C to HDMI Adapter)	0.174110	0.068391	0.020931	0.120214	1.757755	1.0	0.009023	1.058905	0.521973	0.094465	0.055628	0.213129
5	(Anker USB C to HDMI Adapter)	(VIVO Dual LCD Monitor Desk mount)	0.068391	0.174110	0.020931	0.306043	1.757755	1.0	0.009023	1.190117	0.462740	0.094465	0.159746	0.213129
6	(Apple Pencil)	(Apple Lightning to Digital AV Adapter)	0.179709	0.087188	0.028796	0.160237	1.837830	1.0	0.013128	1.086988	0.555754	0.120941	0.080026	0.245256
7	(Apple Lightning to Digital AV Adapter)	(Apple Pencil)	0.087188	0.179709	0.028796	0.330275	1.837830	1.0	0.013128	1.224818	0.499424	0.120941	0.183552	0.245256
8	(Dust-Off Compressed Gas 2 pack)	(Apple Lightning to Digital AV Adapter)	0.238368	0.087188	0.024397	0.102349	1.173883	1.0	0.003614	1.016889	0.194486	0.081009	0.016609	0.191083
9	(Apple Lightning to Digital AV Adapter)	(Dust-Off Compressed Gas 2 pack)	0.087188	0.238368	0.024397	0.279817	1.173883	1.0	0.003614	1.057552	0.162275	0.081009	0.054420	0.191083

Screenshot is somewhat small, please see the attached code on line 18 for a clearer view (or zoom in).

3 and 4 and D1. Here are the top 3 rules for each of support, lift, and confidence.

```
#following code is for showing top 3 rules for each metric
top3 = rulestable.sort_values('confidence', ascending=False).head(3)
top3
```

	antecedents	consequents	antecedent support	consequent support	support	confidence	lift	representativity	leverage	conviction	zhangs_metric	jaccard	certainty	kulczynski
0	(10ft iPHone Charger Cable 2 Pack)	(Dust-Off Compressed Gas 2 pack)	0.050527	0.238368	0.023064	0.456464	1.914955	1.0	0.011020	1.401255	0.503221	0.086760	0.286354	0.276610
37	(FEIYOLD Blue light Blocking Glasses)	(Dust-Off Compressed Gas 2 pack)	0.065858	0.238368	0.027596	0.419028	1.757904	1.0	0.011898	1.310962	0.461536	0.099759	0.237201	0.267400
52	(SanDisk Ultra 64GB card)	(Dust-Off Compressed Gas 2 pack)	0.098254	0.238368	0.040928	0.416554	1.747522	1.0	0.017507	1.305401	0.474369	0.138413	0.233952	0.294127

```
top3 = rulestable.sort_values('lift', ascending=False).head(3)
top3
```

	antecedents	consequents	antecedent support	consequent support	support	confidence	lift	representativity	leverage	conviction	zhangs_metric	jaccard	certainty	kulczynski
85	(VIVO Dual LCD Monitor Desk mount)	(SanDisk Ultra 64GB card)	0.174110	0.098254	0.039195	0.225115	2.291162	1.0	0.022088	1.163716	0.682343	0.168096	0.140684	0.312015
84	(SanDisk Ultra 64GB card)	(VIVO Dual LCD Monitor Desk mount)	0.098254	0.174110	0.039195	0.398915	2.291162	1.0	0.022088	1.373997	0.624943	0.168096	0.272197	0.312015
64	(FEIYOLD Blue light Blocking Glasses)	(VIVO Dual LCD Monitor Desk mount)	0.065858	0.174110	0.022930	0.348178	1.999758	1.0	0.011464	1.267048	0.535186	0.105651	0.210764	0.239939

```
top3 = rulestable.sort_values('support', ascending=False).head(3)
top3
```

	antecedents	consequents	antecedent support	consequent support	support	confidence	lift	representativity	leverage	conviction	zhangs_metric	jaccard	certainty	kulczynski
62	(Dust-Off Compressed Gas 2 pack)	(VIVO Dual LCD Monitor Desk mount)	0.238368	0.174110	0.059725	0.250559	1.439085	1.0	0.018223	1.102008	0.400606	0.169312	0.092566	0.296796
63	(VIVO Dual LCD Monitor Desk mount)	(Dust-Off Compressed Gas 2 pack)	0.174110	0.238368	0.059725	0.343032	1.439085	1.0	0.018223	1.159314	0.369437	0.169312	0.137421	0.296796
41	(Dust-Off Compressed Gas 2 pack)	(HP 61 ink)	0.238368	0.163845	0.052660	0.220917	1.348332	1.0	0.013604	1.073256	0.339197	0.150648	0.068256	0.271158

Screenshot is somewhat small, please see lines 19 thru 21 for a clearer view (or zoom in).

These rules are evaluated using 3 metrics: Support, Confidence, and Lift. Support is a measure of how often an itemset appears in a transaction table. Confidence is a measure of how often an item is purchased assuming that an specific initial item has also been purchased. Lift is a measure of how often items would appear together if they are adjusted for how often the first item appears relative to other items (Doe). To explain using the top rule for each:

Confidence: 10 ft iPhone Charger Cable 2 Pack and Dust-Off Compressed Gas 2 pack – This means that the item most commonly purchased with another item, assuming the first item has been purchased, is the compressed gas 2-pack with the 10ft iPhone charger cable. The confidence value of 0.456464 indicates that when the cable is purchased, there is a 45.6% chance that the customer will also purchase a compressed gas 2-pack. A theoretical explanation is that people often purchase a new charger when their old one stops working, but since they aren't sure if the charger is actually broken or if their charging port is clogged they also purchase the air to de-clog it.

Lift: VIVO Dual LCD Monitor Desk mount and SanDisk Ultra 64GB card – This means that, when adjusted for how often the first item is purchased, the most common pairing is the dual monitor desk mount and the 64GB SD card. The lift value of 2.291162 is well above 1, which indicates a positive association between the two items. It is difficult to explain this one, but perhaps when purchasing peripherals for a computer setup people tend to purchase smaller auxiliary items as well, such as SD cards and compressed gas.

Support: Dust-Off Compressed Gas 2 pack and VIVO Dual LCD Monitor Desk mount – This means that the combination of items that appears most often in the table is the compressed gas 2-pack and the dual monitor desk mount. The support value of 0.059725 indicates 5.97% of all transactions contain these two items. This is probably similar to above, where the person upgrading or purchasing peripherals for their computer is likely cleaning it at the same time or just stocking up on auxiliary items.

Part IV: Data Summary and Implications

D.

1. See above section.

2. Practically speaking, the results of the market basket analysis tend to show a similar takeaway through the top rules; peripheral items, such as the gas 2-pack, 64GB SDcard, and the dual monitor stand are the items most commonly paired with other items. Some instances, such as with the iPhone charging cable, this is easily explainable as needing both items for a similar purpose, which is solving a phone not charging. However, with the gas 2-pack and SD card, the explanation is more likely that these items are simply easy to purchase together with other items and are common-use items.

The MBA findings have good rule-values and thus useful. A company could easily take these findings and use them to discover common associations between items purchased and make practical business-making decisions with them.

3. As the gas 2-pack and SDcard are smaller auxiliary purchases that can be paired with several other item, it would be wise to place these items closer to the checkout/register area at physical locations, or to have them as recommended items in an online cart at checkout. Customers will be more likely to purchase these items than others in such a situation, which would give an easy increase in revenue.

Part V: Attachments

E. See attached link.

F. All code is sourced from Dr. Kamara's videos, or from:

Mlxtend User Guide, https://rasbt.github.io/mlxtend/USER_GUIDE_INDEX/

G. Sources:

Doe, J. (2021, August 15). *Data mining: Market basket analysis with Apriori algorithm*. *Towards Data Science*. https://towardsdatascience.com/data-mining-market-basket-analysis-with-apriori-algorithm-970ff256a92c/

Mlxtend User Guide, https://rasbt.github.io/mlxtend/USER_GUIDE_INDEX/

H. Thank you for reading!